

Requirements that govern AI

Not just describe it.

A step-in-ready framework that turns every requirement into a typed, machine-readable, enforceable engineering artifact — and lets your AI operate inside the control envelope your standards already define.

STAMP / STPA

System-theoretic safety

OWL · SHACL · SWRL

Machine-executable governance

JTBD · Personas · UCAs

Stakeholder-anchored intent

Why every WG keeps hitting the same wall.

Four chronic failures show up in every domain — aerospace, rail, automotive, medical, defense — and in every AI deployment that touches engineering work.

“ Requirements vanish after build

Reqs live in PDFs, DOORS, Jira. The system is built — and the requirements are forgotten. There is no live baseline to trace failures back to.

“ Terminology tribalism kills interop

“Failure mode”, “hazardous event”, “HARA item” — same concept, three vocabularies. STPA classifies this as a formal hazard.

“ AI has no awareness of requirements

Models see raw prompts. They cannot tell you which standard they satisfied, which UCA they avoided, or which persona they served.

“ Governance without enforcement is decoration

A standard you can't automatically evaluate is advice, not control. Silent non-compliance is the default state of most AI deployments today.

The RDSG addresses all four — by architecture, not by discipline.

From management to engineering of requirements.

REQUIREMENTS MANAGEMENT

Today's default

- Documents stored in tools, then forgotten
- No machine-readable structure
- Traceability is manual and brittle
- Ambiguity managed by convention
- AI has no awareness of requirements



REQUIREMENTS ENGINEERING

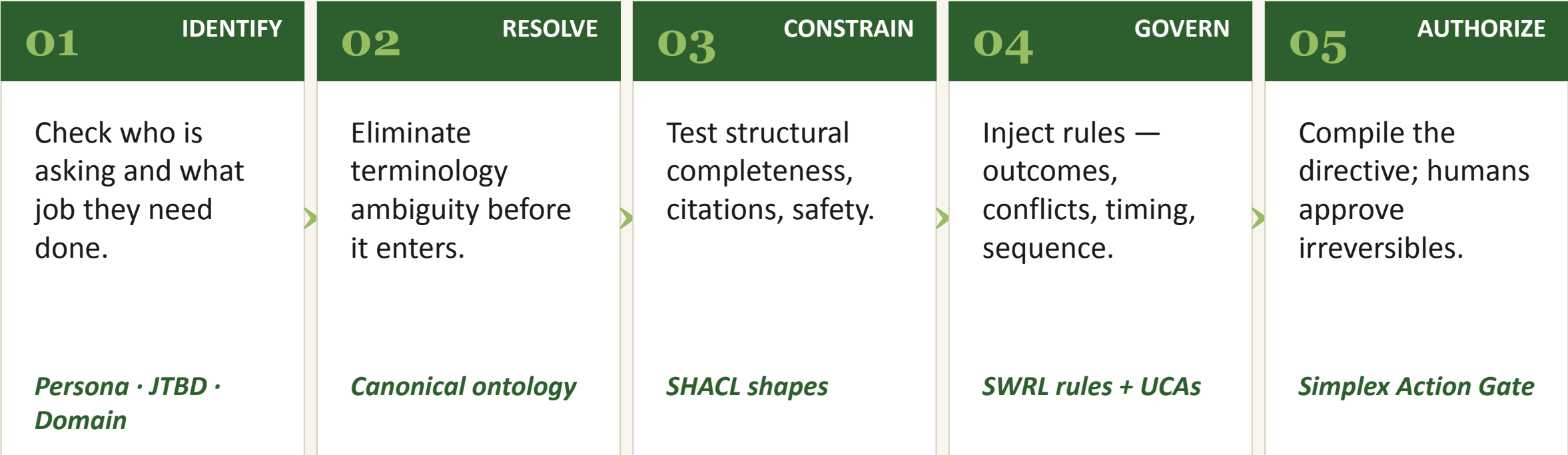
What the RDSG enables

- ✓ Reqs are typed ontology individuals in a triplestore
- ✓ Machine-readable, formally typed, versioned
- ✓ Full traceability — automated and queryable
- ✓ Ambiguity detected and resolved at the gateway
- ✓ AI operates inside a requirement-defined envelope

If a requirement only exists in a PDF, it does not exist.

The gateway — five steps, plain language.

Every AI interaction passes through five governed steps. The AI can only respond inside the envelope these steps define.



SERVICE FAÇADE · 10-criterion validation

Every AI response is evaluated against acceptance criteria before it ever reaches a human.

SIMPLEX ACTION GATEWAY

No irreversible action without a verified human handshake.

Where every WG expertise plugs in.

Controllability over reliability

AI shifts from a stochastic risk to a constrained control problem — STAMP-aligned, scalable as capabilities grow.

4 / 4 UCA coverage

Missing · Hazardous · Mis-timed · Wrong-duration. STPA's complete failure-mode taxonomy, mapped to executable rules.

End-to-end traceability

Stakeholder need → ontology individual → SHACL shape → SWRL rule → CAD → validated response. Queryable on demand.

WHERE YOU PLUG IN

Every WG expertise has a defined point of contribution — no ontology background required.

Domain practitioner

AEROSPACE · RAIL · AUTO

Bring the requirements, the constraints, the failure modes.

Systems engineer

MBSE · ARCHITECTURE

Map JTBD hierarchies to system decomposition; UCAs to design constraints.

Safety analyst

STPA · HAZOP · FMEA

Define UCAs per job; classify Zone 6 actions by reversibility.

Ontologist

KNOWLEDGE ENG.

Author canonical vocabulary; design SHACL shapes and SWRL rules.

Compliance steward

POLICY · STANDARDS

Activate, validate, retire requirements; trigger reviews.

ASK → Bring 5 contested terms · 1 real requirement · 1 interoperability failure · 1 SHACL shape from your domain.